

AZURE MONITOR ALERT, ACTION GROUP AND ON-CALL ROUTING

Project Documentation

Project Domain: Cloud Monitoring and Alert Automation

Resource Group: rg-monitor-project

1. Executive Summary

The **Azure Monitor Alert, Action Group and On-Call Routing** project is a cloud monitoring solution designed to continuously monitor the health and performance of an Azure Virtual Machine.

The project uses **Azure Monitor** to observe CPU utilization and identify abnormal resource usage. When CPU utilization exceeds the configured threshold, an alert is automatically generated. The alert is connected to an **Azure Monitor Action Group**, which sends a notification to the responsible on-call team through email.

The project demonstrates an automated monitoring and notification workflow rather than relying on manual monitoring. It also includes a controlled testing process in which CPU load is increased to trigger the alert and is later reduced to verify that the alert is resolved correctly.

The implementation uses Azure resources including a Virtual Machine, Azure Monitor, Log Analytics Workspace, Data Collection Rule, Alert Rule, and Action Groups.

2. Project Overview

Cloud applications and infrastructure must be continuously monitored to ensure that resources are operating within acceptable limits. A sudden increase in CPU utilization can indicate a heavy workload or an operational issue that requires attention.

In this project, an Azure Virtual Machine named **monitor-vm** is monitored using Azure monitoring services. A CPU alert named **CPU-High-T065** is configured to detect CPU utilization above **80%**.

When the condition is satisfied:

Virtual Machine → Azure Monitor → Alert Rule → Action Group → Email Notification → On-Call Team

This provides a simple automated mechanism for detecting and communicating infrastructure issues.

3. Problem Statement

Manual monitoring of cloud resources can be inefficient because administrators cannot continuously observe system performance.

If CPU utilization becomes unusually high and the condition is not identified quickly, it may affect the performance and availability of applications running on the virtual machine.

Therefore, the project aims to implement an automated monitoring system that:

- Continuously monitors the virtual machine.
- Detects high CPU utilization.
- Automatically generates an alert.
- Routes the alert to the appropriate team.
- Sends an email notification.
- Tracks the recovery and resolution of the alert.

4. Project Objectives

The main objectives of the project are:

1. To monitor an Azure Virtual Machine using Azure Monitor.
2. To configure CPU utilization monitoring.
3. To create an alert rule for high CPU usage.
4. To configure an appropriate CPU threshold.
5. To create Action Groups for team-based notification.

6. To send automated email notifications when an alert is triggered.
7. To test the alert mechanism using controlled CPU load.
8. To verify successful notification delivery.
9. To verify that the alert returns to a resolved state after recovery.
10. To understand the basic principles of cloud monitoring and incident notification.

5. Project Architecture

The project follows a simple monitoring and notification architecture.

Main Architecture

Azure Virtual Machine

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Azure Monitor

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CPU Alert Rule

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Action Group

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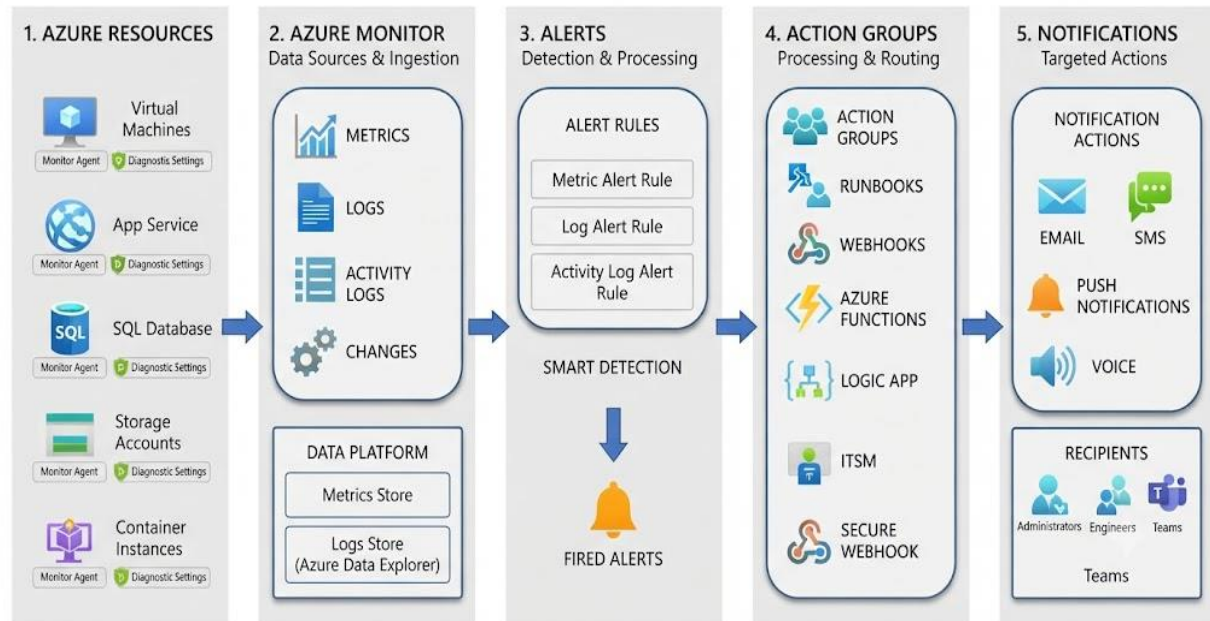
Email Notification

↓

On-Call Team

The Virtual Machine generates monitoring data that is evaluated by Azure Monitor. When the configured CPU condition is satisfied, the alert rule is activated. Azure Monitor then invokes the associated Action Group, which performs the configured notification action.

AZURE MONITOR ARCHI ARCHITECTURE: END-TO-END FLOW



6. Azure Resources Used

Azure Resource	Resource Name	Purpose
Resource Group	rg-monitor-project	Organizes project resources
Virtual Machine	monitor-vm	Primary monitored resource
Log Analytics Workspace	law-p5xsjcyuoybns	Monitoring and log analysis
Data Collection Rule	monitor-vm-dcr	Defines telemetry collection

Alert Rule	CPU-High-T065	Detects high CPU utilization
Action Group	AG-T065	Notification routing for Team T065
Action Group	AG-T167	Notification routing for Team T167
Action Group	AG-T269	Notification routing for Team T269

The VM is monitored for performance-related conditions, with CPU utilization selected as the primary metric for demonstrating automated alerting.

The VM acts as the starting point of the monitoring workflow. Its performance data is evaluated by Azure monitoring services, allowing abnormal CPU utilization to be detected automatically.

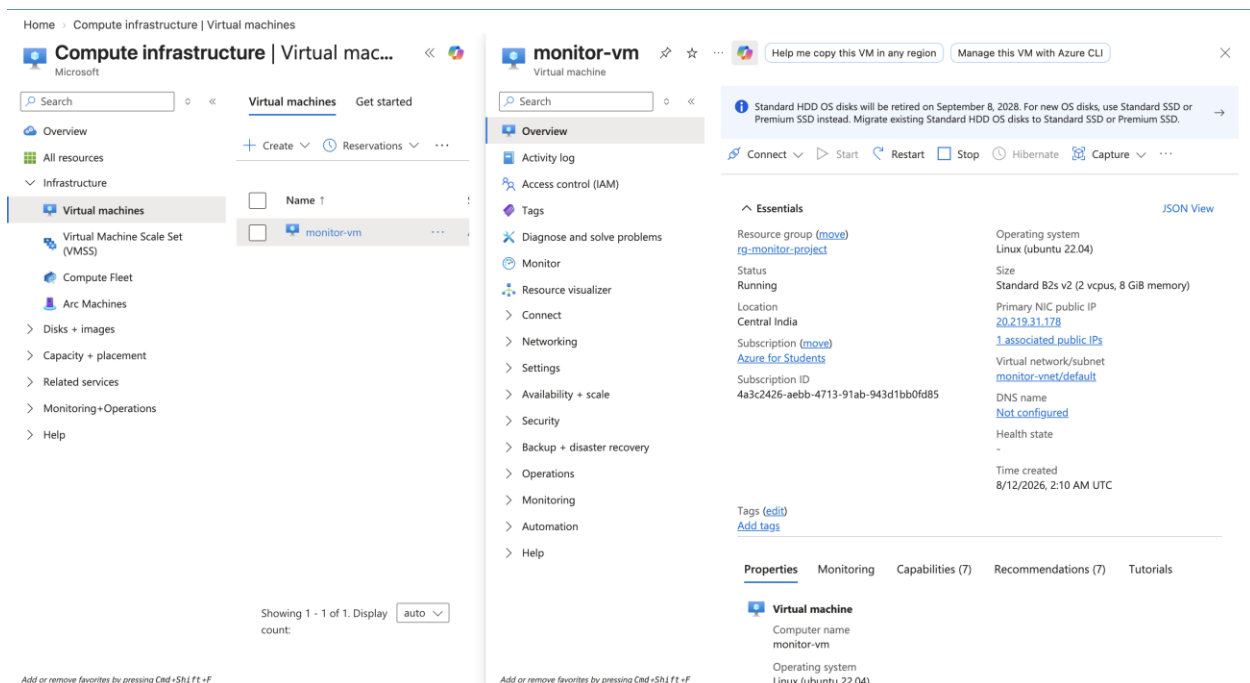
Monitoring Focus

Metric: CPU Utilization

Alert Condition: CPU utilization above 80%

Alert: CPU-High-T065

Severity: 0



8. Azure Monitor

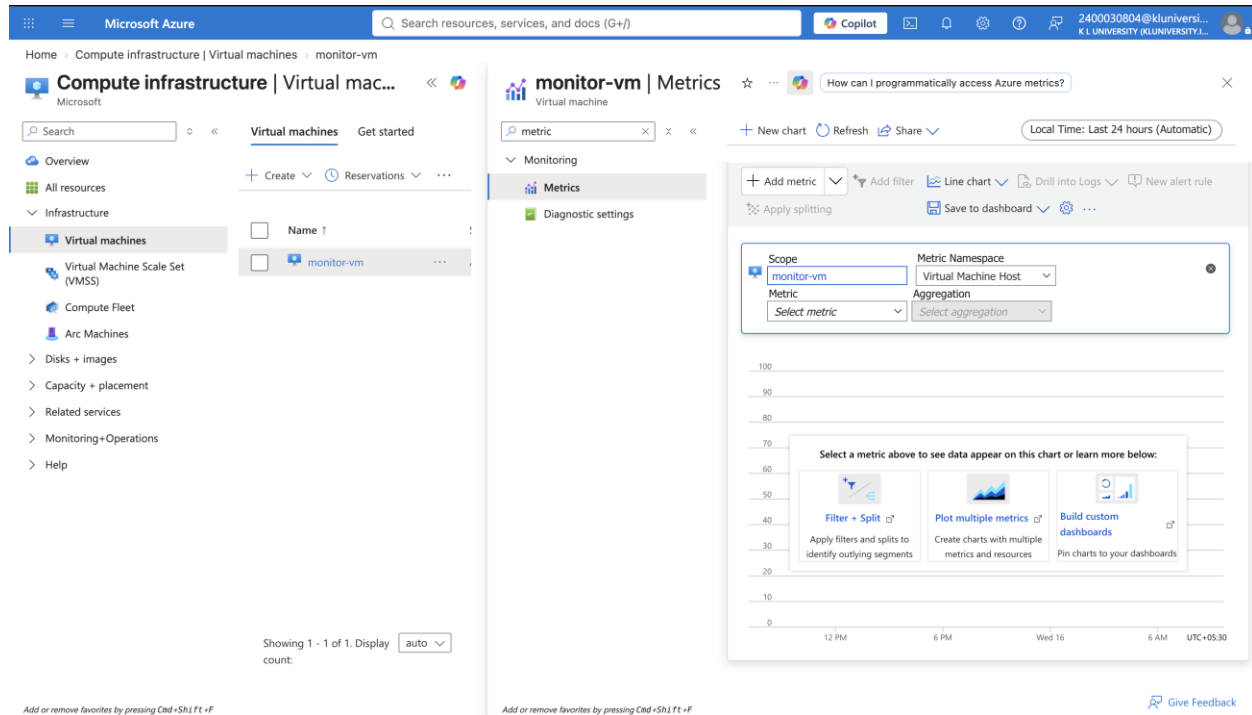
Azure Monitor is the central monitoring service used in the project.

It collects and analyzes monitoring information from Azure resources and allows conditions to be defined for detecting abnormal behavior.

In this project, Azure Monitor evaluates CPU utilization of the virtual machine. When the configured condition is met, it activates the corresponding alert.

The monitoring process can be represented as:

Collect → Evaluate → Detect → Alert → Notify → Resolve



9. CPU Alert Configuration

A dedicated alert rule named **CPU-High-T065** is configured for the project.

The purpose of this alert is to identify situations where the CPU utilization of the monitored virtual machine exceeds the defined threshold.

Alert Configuration

Parameter	Configuration
Alert Name	CPU-High-T065
Monitored Resource	monitor-vm
Metric	CPU Utilization
Threshold	Greater than 80%
Severity	0
Action	Action Group notification

When CPU utilization crosses the configured threshold, Azure Monitor changes the alert into an active state and triggers the associated notification mechanism.

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Home > Monitor | Alerts

Alert rules

CreateColumnsRefreshExport to CSVOpen queryDeleteEnableDisable

Create an autonomous observability agent: Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#) Create agent

Search

Subscription : Azure for StudentsTarget scope : allTarget resource type : allSignal type : allSeverity : allAdd tag filterMore (1)No grouping

Name	Condition	Severity	Target scope	Target resource type	Signal type	Status
☐ CPU-High-T065	Percentage CPU > 80	0 - Critical	monitor-vm	Virtual machine	Metrics	Enabled

Showing 1 - 1 of 1 results.

Give feedback

10. Action Groups

Azure Monitor **Action Groups** provide a mechanism for defining what should happen when an alert is triggered.

In this project, separate Action Groups have been configured for the project teams:

- **AG-T065**
- **AG-T167**
- **AG-T269**

The Action Groups contain the notification configuration required to inform the responsible team when a monitoring condition occurs.

Routing Model

Alert Triggered

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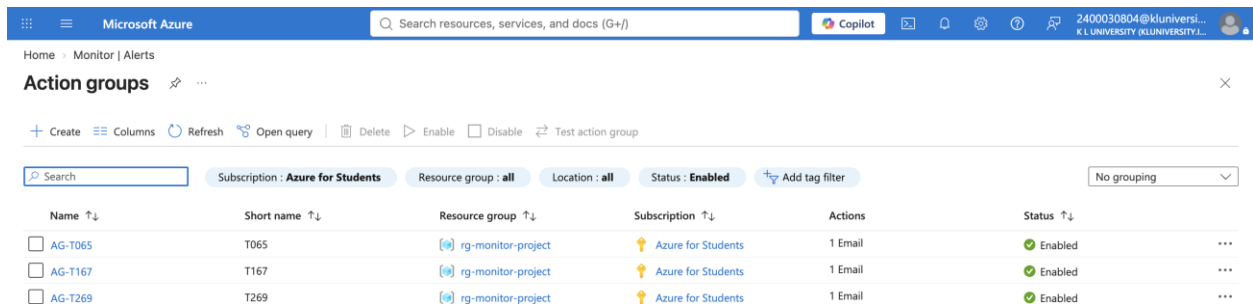
Action Group Activated

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Configured Notification

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On-Call Team Receives Notification



Name ↑↓	Short name ↑↓	Resource group ↑↓	Subscription ↑↓	Actions	Status ↑↓
☐ AG-T065	T065	rg-monitor-project	Azure for Students	1 Email	Enabled
☐ AG-T167	T167	rg-monitor-project	Azure for Students	1 Email	Enabled
☐ AG-T269	T269	rg-monitor-project	Azure for Students	1 Email	Enabled

11. On-Call Notification Workflow

The project implements an automated on-call notification workflow.

When the CPU utilization of the monitored VM exceeds 80%, the following sequence takes place:

Step 1 – Monitoring

Azure Monitor continuously evaluates the selected monitoring signal.

Step 2 – Threshold Detection

The CPU utilization exceeds the configured threshold of 80%.

Step 3 – Alert Generation

The CPU-High-T065 alert rule is activated.

Step 4 – Action Group Execution

The configured Action Group is executed.

Step 5 – Notification

An email notification is sent to the configured recipient.

Step 6 – Incident Response

The responsible team can investigate the condition and take appropriate action.

Step 7 – Recovery

When CPU utilization returns to a normal state, the alert can transition to a resolved state.

Microsoft Azure

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Home > Monitor | Alerts > Action groups

Action groups

+ Create Columns ...

Search

Name ↑↓

- AG-T065
- AG-T167
- AG-T269

AG-T065
Action group

How do I troubleshoot issues with this Action group?

Search Edit Disable Test Delete Refresh

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Automation
- Help

Essentials

Resource group [\(move\)](#)
[rg_monitor-project](#)

Location [\(move\)](#)
Global

Subscription [\(move\)](#)
[Azure for Students](#)

Subscription ID
4A3C2426-AEBB-4713-91AB-943D1BB0FD85

Tags [\(edit\)](#)
[Add tags](#)

Display name
T065

JSON View

Notifications [Resend verification emails](#)

Type	Name	Value	Email verification
Email	T065	2400030804@kluniv...	Legacy allowed

AZURE MONITOR ALERT FLOW



12. Monitoring Lifecycle

The complete monitoring lifecycle can be represented as:

Telemetry Collection

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Metric Evaluation

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Threshold Detection

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Alert Generation

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Action Group Execution

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Email Notification

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Team Response

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Resource Recovery

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Alert Resolution

This lifecycle demonstrates how cloud monitoring can support faster identification and communication of infrastructure issues.

AZURE VM ALERT & RESOLUTION WORKFLOW



13. Log Analytics Workspace

The project uses the Log Analytics Workspace:

law-p5xsjcyuoybns

A Log Analytics Workspace provides a centralized environment for collecting and analyzing monitoring and log data.

It supports the broader monitoring architecture by providing a location where relevant telemetry and operational information can be analyzed.

14. Data Collection Rule

The project uses the Data Collection Rule:

monitor-vm-dcr

A Data Collection Rule defines how monitoring data is collected and processed.

Within the project architecture, the Data Collection Rule forms part of the telemetry collection layer and supports the monitoring configuration associated with the virtual machine.

Recommended Image

Insert screenshot: Data Collection Rule → monitor-vm-dcr

15. Implementation Process

The implementation was carried out through the following major stages:

Phase 1 – Resource Setup

The required Azure resources were organized under the resource group:

rg-monitor-project

Phase 2 – Virtual Machine Configuration

The Azure Virtual Machine **monitor-vm** was prepared as the monitored resource.

Phase 3 – Monitoring Configuration

Azure Monitor and supporting monitoring resources were configured.

Phase 4 – Telemetry Configuration

The Log Analytics Workspace and Data Collection Rule were included in the monitoring setup.

Phase 5 – Action Group Configuration

Team-specific Action Groups were created for T065, T167 and T269.

Phase 6 – Alert Creation

The CPU alert **CPU-High-T065** was created with a threshold of greater than 80%.

Phase 7 – Notification Testing

A controlled CPU load was generated on the VM to test the alert mechanism.

Phase 8 – Recovery Testing

The CPU load was stopped and the alert state was monitored until it returned to normal.

16. Testing and Validation

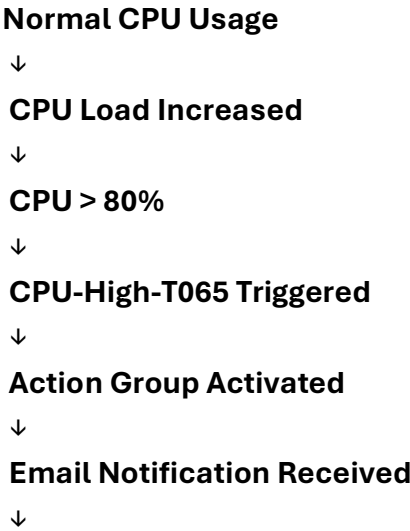
Testing was performed to verify the complete monitoring and notification workflow.

Test Case	Test Condition	Expected Result	Result
TC-01	CPU below threshold	No high-CPU alert	Normal monitoring
TC-02	CPU exceeds 80%	Alert is triggered	Alert triggered
TC-03	Alert becomes active	Email notification is sent	Notification verified
TC-04	CPU returns to normal	Alert is resolved	Resolution verified

The controlled CPU load test provided a practical demonstration that the configured alert could detect the specified condition.

17. Alert Triggering Scenario

The alert testing process can be represented as:



CPU Load Stopped

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CPU Returns to Normal

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Alert Resolved

18. Results

The implementation successfully demonstrated the core functionality of the proposed monitoring system.

The project achieved the following:

- Azure VM monitoring was configured.
- CPU utilization was selected as the primary monitoring metric.
- A CPU threshold of greater than 80% was configured.
- The CPU-High-T065 alert was successfully tested.
- The Action Group notification mechanism was configured.
- Email notification was verified during testing.
- The alert returned to a resolved state after CPU utilization returned to normal.

The project therefore demonstrates a complete basic workflow from **resource monitoring to automated notification and alert resolution**.

19. Security and Reliability Considerations

Security and reliability are important aspects of any cloud monitoring implementation.

The following practices should be considered:

- Apply least-privilege access to Azure resources.
- Restrict monitoring configuration permissions to authorized users.
- Avoid storing passwords, access keys, or other secrets in project documentation.
- Use clear and consistent resource names.
- Ensure notification recipients are correctly configured.

- Regularly review alert thresholds.
- Test alert rules after major configuration changes.
- Monitor the monitoring system itself to reduce the possibility of missed alerts.

20. Project Benefits

The implemented solution provides several operational benefits:

Automated Monitoring

The system reduces the need for continuous manual observation of the VM.

Faster Notification

When the defined condition occurs, the responsible team receives a notification automatically.

Centralized Monitoring

Azure Monitor provides a central platform for observing Azure resources.

Team-Based Routing

Action Groups allow notifications to be associated with specific project teams.

Testable Workflow

The controlled CPU load test provides a practical way to validate the monitoring configuration.

Extensibility

The same architecture can be extended to monitor additional infrastructure metrics.

21. Conclusion

The **Azure Monitor Alert, Action Group and On-Call Routing** project demonstrates how Azure cloud monitoring services can be used to automatically detect infrastructure conditions and communicate them to responsible teams.

The solution combines a monitored Azure Virtual Machine, Azure Monitor, telemetry collection, an alert rule, Action Groups, and email notification into a single workflow.

The successful CPU load test demonstrates the complete lifecycle of the solution: **monitoring, threshold detection, alert generation, notification, and resolution.**

The project provides a practical foundation for developing more advanced cloud monitoring and automated incident-response systems in the future.